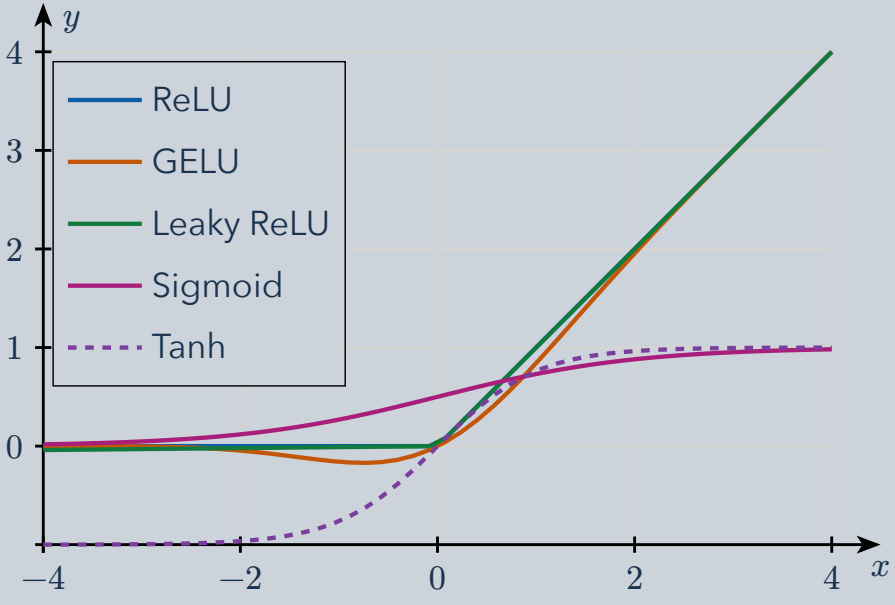
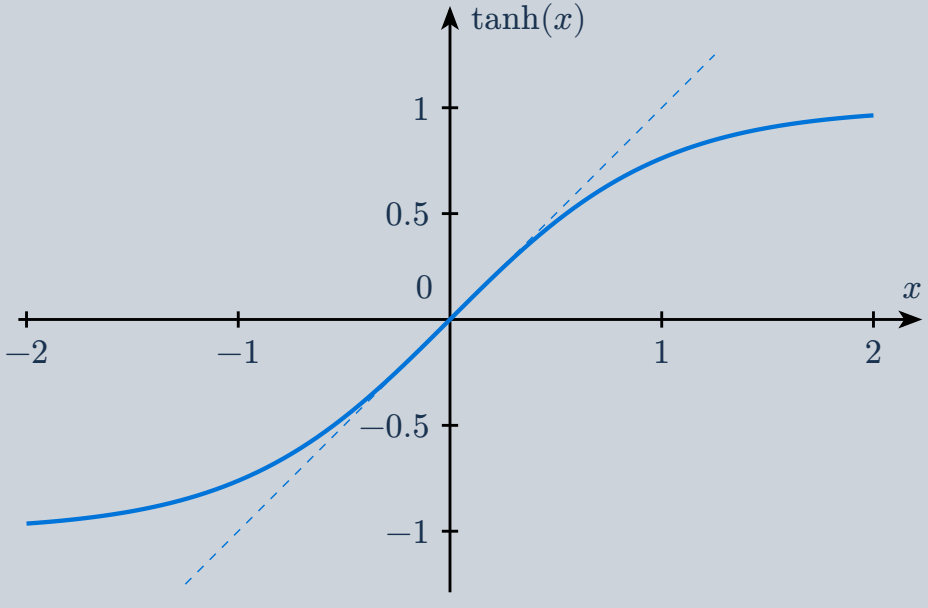
Activation functions turn a linear network into a nonlinear model. Compare their outputs, then inspect how tanh trades a linear center for saturation at the edges.

1 Common nonlinearities



ReLU clips negative inputs; leaky ReLU keeps a small negative slope. GELU gates smoothly. Sigmoid maps to $(0, 1)$ and tanh to $(-1, 1)$.

2 Tanh: linear center, saturated tails



Near zero, $\tanh x \approx x$. Far from zero, the output approaches $+1$ or -1 and its slope approaches zero.

$\frac{\mathrm{d}}{\mathrm{d}x} \tanh x = 1 - \tanh^2 x$. **Small slope means a small backpropagated gradient.** Saturation can bound an output but also make learning through that unit slow.